

Qatar-4b

R-2690

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_200726 · created 2026-08-31T01:20:45+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459056.7883 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.047
Transit depth (Rp/Rs)^2	1.8 +/- 1.27 [%]
Orbital Inclination (inc)	87.6 +/- 2.09
Airmass coefficient 1 (a1)	0.971 +/- 0.033
Airmass coefficient 2 (a2)	0.02 +/- 0.022
Scatter in the residuals of the lightcurve fit is	3.38 %
Best Comparison Star	#2 - [407, 255]
Optimal Aperture	3.08
Optimal Annulus	10.52
Transit Duration (day)	0.0753 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.134 $\pm$ 0.047 vs 0.138 (deviation 0.09 σ)
Duration fit vs published	0.0753 d vs 0.0892 d (deviation 2.57 σ)
Mid-time O-C	-4.2 min
Depth SNR	2.9
Residual scatter	3.38 %

Light curve

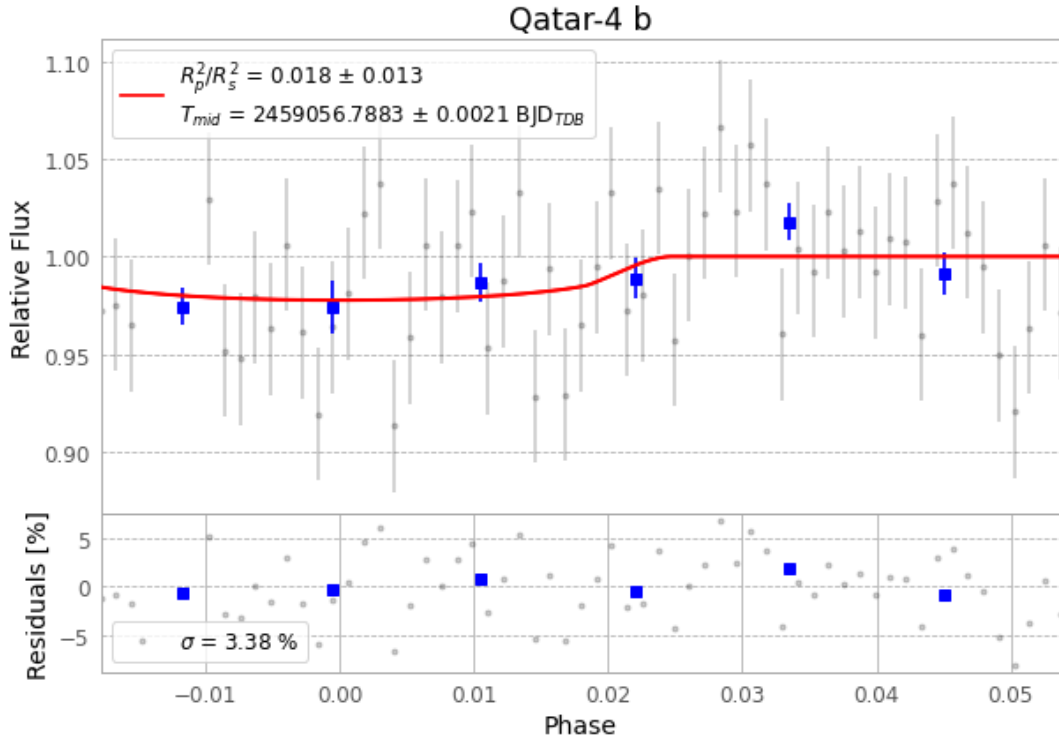


Figure 1 — FinalLightCurve_Qatar-4 b_2020-07-25.png (SHA-256 08c195220880bfa9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [447, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01983475798b828f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

63 FITS frames (41 MB), Qatar-4200726060612.FITS → Qatar-4200726091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-200726.log (653ba4127087...), FinalLightCurve_Qatar-4 b_2020-07-25.png (08c195220880...), FinalLightCurve_Qatar-4 b_2020-07-25.csv (92de236a127e...)

Compute resources

Wall time 143.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-31 01:20Z.